

Lecture 4

Canonical and Standard Forms

Minterms and maxterms · canonical forms · SOP and POS · the 16 two-variable functions · the standard gates

xyz	F
000	0
001	1
010	1
011	0
100	1
101	0
110	0
111	1

$$F = \Sigma m(1,2,4,7)$$

Reading: Mano & Ciletti, *Digital Design with an Introduction to the Verilog HDL*, 6th ed. — Chapter 2, §2.6–2.8

Today's roadmap

1

§2.6

Canonical forms

Minterms and maxterms

2

§2.6

Standard forms

SOP, POS, two levels

3

§2.7

16 functions

All two-input operations

4

§2.8

Logic gates

Eight gates, many inputs

By the end of class you can...

- Write any function as a sum of minterms or product of maxterms
- Convert between Σ and Π forms
- Draw SOP and POS circuits and count levels
- Name the 16 functions of two variables
- Use multiple-input gates and positive or negative logic

Agenda for today's session

Intro

Canonical

Standard

Break

16 funcs

Gates

Wrap

0:00

0:05

0:35

0:50

1:00

1:15

1:40

1:50

Elapsed time	Topic	In-class work
0:00 – 0:05	Welcome, roadmap	—
0:05 – 0:35	Minterms, maxterms, conversion	Examples 1, 2 · Your turn 1
0:35 – 0:50	SOP, POS, two and three levels	Example 3
0:50 – 1:00	Break	
1:00 – 1:15	Other logic operations	—
1:15 – 1:40	Standard gates, multi-input, pos./neg. logic	Example 4 · Your turn 2
1:40 – 1:50	Concept check and wrap-up	Homework preview

Minterms and maxterms

§2.6

x y z	Minterm		Maxterm	
0 0 0	$x'y'z'$	m0	$x+y+z$	M0
0 0 1	$x'y'z$	m1	$x+y+z'$	M1
0 1 0	$x'yz'$	m2	$x+y'+z$	M2
0 1 1	$x'yz$	m3	$x+y'+z'$	M3
1 0 0	$xy'z'$	m4	$x'+y+z$	M4
1 0 1	$xy'z$	m5	$x'+y+z'$	M5
1 1 0	xyz'	m6	$x'+y'+z$	M6
1 1 1	xyz	m7	$x'+y'+z'$	M7

Minterm m_j

AND of all variables: primed where the bit is 0. It is 1 on row j only.

Maxterm M_j

OR of all variables: primed where the bit is 1. It is 0 on row j only.

Each is the other's complement

$$m'_j = M_j$$

Example: $m_5 = xy'z$, $M_5 = x' + y + z'$

Canonical forms

§2.6

x y z	f ₁	f ₂
0 0 0	0	0
0 0 1	1	0
0 1 0	0	0
0 1 1	0	1
1 0 0	1	0
1 0 1	0	1
1 1 0	0	1
1 1 1	1	1

Sum of minterms: OR the rows where $f = 1$

$$f_1 = x'y'z + xy'z' + xyz = m_1 + m_4 + m_7 = \Sigma m(1,4,7)$$

$$f_2 = x'yz + xy'z + xyz' + xyz = \Sigma m(3,5,6,7)$$

Product of maxterms: AND the rows where $f = 0$

$$f_1 = M_0M_2M_3M_5M_6 = \Pi M(0,2,3,5,6)$$

$$f_2 = (x + y + z)(x + y + z')(x + y' + z)(x' + y + z) = \Pi M(0,1,2,4)$$

Converting: Σ and Π use complementary index lists.

Example 1 — Expand to minterms

§2.6

EXAMPLE

Express $F = A + B'C$ as a sum of minterms, algebraically and by truth table.

A B C	A	B'C	F
0 0 0	0	0	0
0 0 1	0	1	1
0 1 0	0	0	0
0 1 1	0	0	0
1 0 0	1	0	1
1 0 1	1	1	1
1 1 0	1	0	1
1 1 1	1	0	1

- 1 Term A is missing B and C : $A = A(B + B')(C + C')$
- 2 Multiply out: $ABC + ABC' + AB'C + AB'C'$
- 3 Term $B'C$ is missing A : $B'C(A + A') = AB'C + A'B'C$
- 4 $AB'C$ appears twice: keep one ($x + x = x$)
- 5 $F = A'B'C + AB'C' + AB'C + ABC' + ABC$
- 6 Number each term: $001 = 1, 100 = 4, 101 = 5, 110 = 6, 111 = 7$
- 7
$$F = \Sigma m(1,4,5,6,7)$$
- 8 The table has 1s on the same five rows ✓

Example 2 — Expand to maxterms

§2.6

EXAMPLE

Express $F = xy + x'z$ as a product of maxterms.

Procedure

1. Use $x + yz = (x + y)(x + z)$ to reach OR terms.
2. Drop any term equal to 1.
3. Add a missing variable with $+xx'$ and distribute.

Check with the complement list: $F = \Sigma m(1,3,6,7)$, so $F = \Pi M(0,2,4,5)$.

- 1 Distribute + over $\cdot$: $F = (xy + x')(xy + z)$
- 2 Again: $(x + x')(y + x')(x + z)(y + z)$
- 3 $x + x' = 1$: $F = (x' + y)(x + z)(y + z)$
- 4 $x' + y$ lacks z : $(x' + y + z)(x' + y + z') = M_4M_5$
- 5 $x + z$ lacks y : $(x + y + z)(x + y' + z) = M_0M_2$
- 6 $y + z$ lacks x : $(x + y + z)(x' + y + z) = M_0M_4$
- 7 Collect, dropping repeats: $M_0M_2M_4M_5$

8

$$F = \Pi M(0,2,4,5)$$

Your turn 1 — Canonical forms

§2.6

YOUR TURN

≈ 10 min

(a) $F(A, B, C) = \Sigma m(1, 4, 5, 6, 7)$. Write F' as a sum of minterms and F as a product of maxterms.

(b) Write $F = A'C + AB'$ as a sum of minterms and as a product of maxterms.

(c) Expand $(A + C)$ over the variables A, B, C into maxterms.

Use a truth table to check (b).

Your turn 1 — Solution

§2.6

SOLUTION

1 (a) Minterms not in F : $F' = \Sigma m(0,2,3)$

2 $F = (F')' = (m_0 + m_2 + m_3)'$ (involution)

3 DeMorgan: $m'_0 m'_2 m'_3$

4 $m'_j = M_j$: $F = \Pi M(0,2,3)$

5 (b) $A'C(B + B') = A'BC + A'B'C = m_3 + m_1$

6 $AB'(C + C') = AB'C + AB'C' = m_5 + m_4$

7 $F = \Sigma m(1,3,4,5)$

8 Missing indices 0, 2, 6, 7: $F = \Pi M(0,2,6,7)$

9 (c) $A + C$ lacks B : $A + C + BB'$

10 Distribute: $(A + B + C)(A + B' + C)$

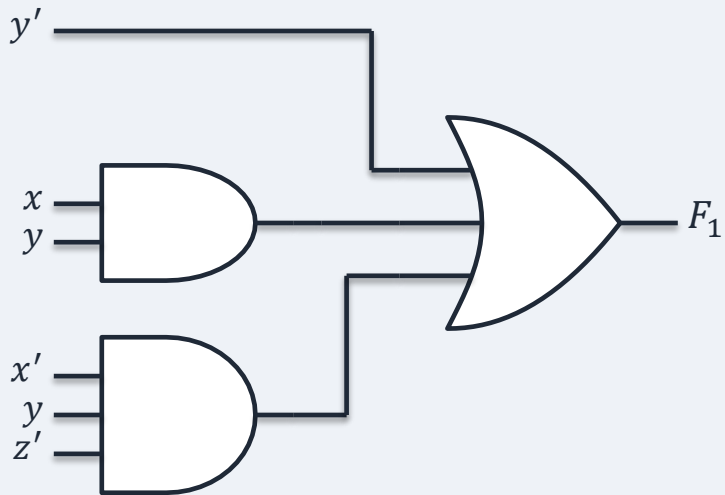
11 $A + B + C = M_0$, $A + B' + C = M_2$

12 $A + C = M_0 M_2$

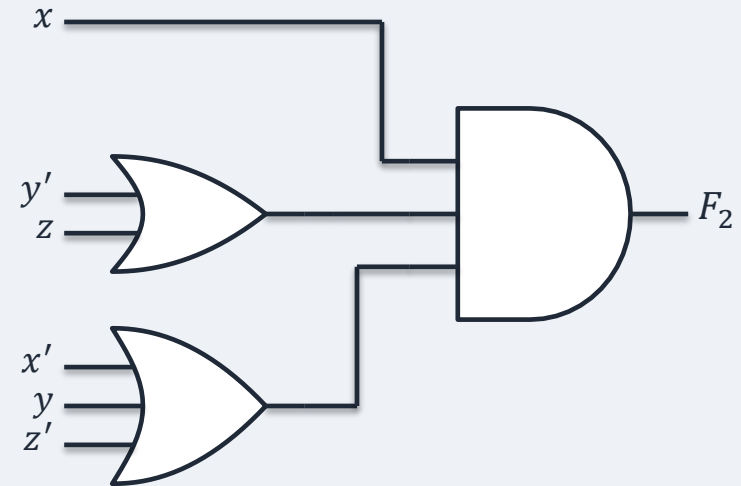
Standard forms: SOP and POS

§2.6

Sum of products: $F_1 = y' + xy + x'yz'$



Product of sums: $F_2 = x(y' + z)(x' + y + z')$



Standard forms need not contain every variable in each term, so they are usually smaller than canonical forms. Both are two-level: AND-OR and OR-AND.

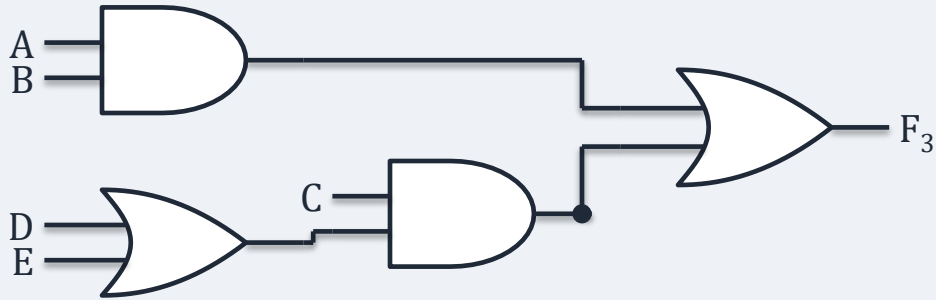
Example 3 — Two levels or three?

§2.6

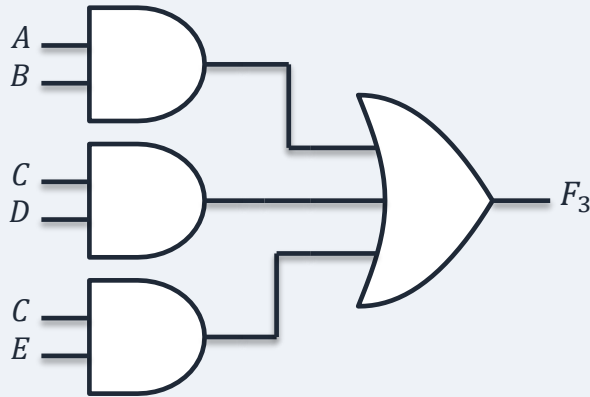
EXAMPLE

Compare $F_3 = AB + C(D + E)$ with its two-level form. Assume a delay t per gate.

Three levels: $AB + C(D + E)$



Two levels: $AB + CD + CE$



- 1 Distribute: $C(D + E) = CD + CE$, so $F_3 = AB + CD + CE$
- 2 Three-level: longest path $E \rightarrow \text{OR} \rightarrow \text{AND} \rightarrow \text{OR} = 3t$
- 3 Two-level: every path is $\text{AND} \rightarrow \text{OR} = 2t$
- 4 Gate inputs, three-level: $2 + 2 + 2 + 2 = 8$
- 5 Gate inputs, two-level: $2 + 2 + 2 + 3 = 9$
- 6 Check $A = B = 0, C = 1, D = 0, E = 1$: $0 + 1(0 + 1) = 1$
- 7 and $0 + 0 + 1 = 1 \checkmark$

Break

10 minutes — we resume at the 1:00 mark

Next: the 16 functions of two variables and the standard gates

The 16 functions of two variables

§2.7

F	00 01 10 11	Expression	Name
F0	0 0 0 0	0	Null
F1	0 0 0 1	xy	AND
F2	0 0 1 0	xy'	Inhibition
F3	0 0 1 1	x	Transfer
F4	0 1 0 0	$x'y$	Inhibition
F5	0 1 0 1	y	Transfer
F6	0 1 1 0	$x \oplus y$	Exclusive-OR
F7	0 1 1 1	$x+y$	OR

F	00 01 10 11	Expression	Name
F8	1 0 0 0	$(x+y)'$	NOR
F9	1 0 0 1	$(x \oplus y)'$	Equivalence
F10	1 0 1 0	y'	Complement
F11	1 0 1 1	$x+y'$	Implication
F12	1 1 0 0	x'	Complement
F13	1 1 0 1	$x'+y$	Implication
F14	1 1 1 0	$(xy)'$	NAND
F15	1 1 1 1	1	Identity

The column for $xy = 00,01,10,11$ read as a binary number is the subscript. Highlighted: the gates we build.

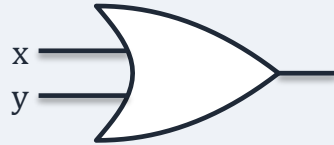
The eight standard gates

§2.8



AND

$$xy \quad (x = 1, y = 0: 0)$$



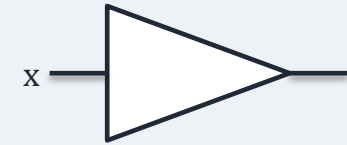
OR

$$x + y \quad (x = 1, y = 0: 1)$$



NOT

$$x' \quad (x = 1, y = 0: 0)$$



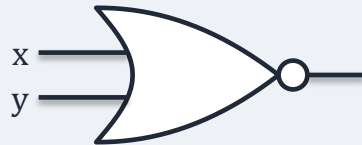
Buffer

$$x \quad (x = 1, y = 0: 1)$$



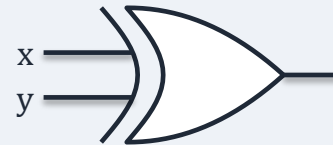
NAND

$$(xy)' \quad (x = 1, y = 0: 1)$$



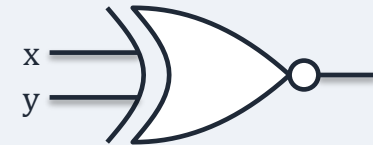
NOR

$$(x + y)' \quad (x = 1, y = 0: 0)$$



XOR

$$x \oplus y \quad (x = 1, y = 0: 1)$$



XNOR

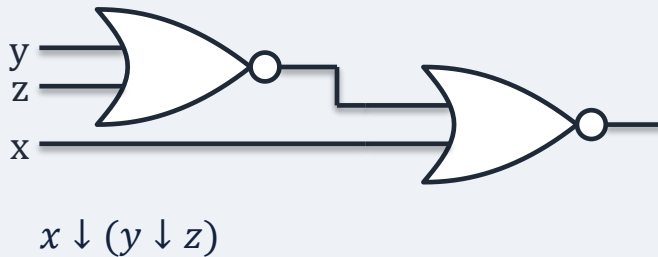
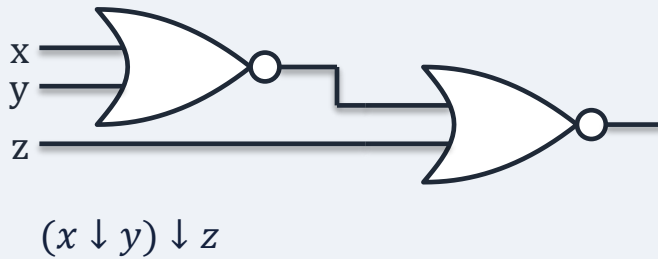
$$(x \oplus y)' \quad (x = 1, y = 0: 0)$$

Example 4 — NOR is not associative

§2.8

EXAMPLE

Show that $(x \downarrow y) \downarrow z \neq x \downarrow (y \downarrow z)$, where $\downarrow$ is NOR.

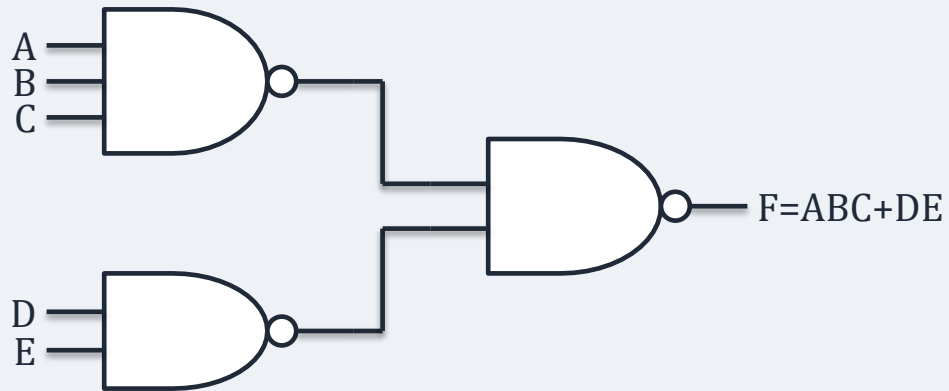


- 1 Inner NOR: $x \downarrow y = (x + y)'$
- 2 Outer: $[(x + y)' + z]' = (x + y)''z' = (x + y)z'$ (DeMorgan)
- 3 Other cascade: $[x + (y + z)']' = x'(y + z)'' = x'(y + z)$
- 4 Try $x = 1, y = 0, z = 0$ in the first: $(1 + 0) \cdot 1 = 1$
- 5 and in the second: $0 \cdot (0 + 0) = 0$
- 6 The results differ: **not associative**
- 7 So a 3-input NOR is defined as $(x + y + z)'$.

Multiple-input NAND and XOR

§2.8

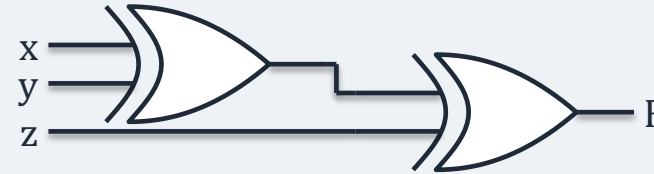
Cascaded NAND: $F = [(ABC)'(DE)']'$



DeMorgan: $F = (ABC)'' + (DE)'' = ABC + DE$

Two levels of NAND build any SOP (Lecture 7).

Three-input XOR: an odd function



x y z	F
0 0 0	0
0 0 1	1
0 1 0	1
0 1 1	0
1 0 0	1
1 0 1	0
1 1 0	0
1 1 1	1

$F = 1$ when an odd number of inputs is 1.

$$F = \sum m(1,2,4,7)$$

XOR is associative, so two 2-input gates in any order work.

Positive and negative logic

§2.8

x	y	z
L	L	L
L	H	L
H	L	L
H	H	H

Physical gate: voltages H and L

x	y	z
0	0	0
0	1	0
1	0	0
1	1	1

Positive (H = 1): AND

x	y	z
1	1	1
1	0	1
0	1	1
0	0	0

Negative (H = 0): OR

The rule

Switching every signal to negative logic turns a function into its dual.

AND $\leftrightarrow$ OR, NAND $\leftrightarrow$ NOR

Small triangles on the pins (polarity indicators) mark negative-logic signals.

A positive-logic AND chip is a negative-logic OR chip: the hardware is the same, only the labels change.

Your turn 2 — Draw and simplify

§2.8

YOUR TURN

≈ 10 min

(a) Draw $F = D + BC + (D + C')(A' + C)$ exactly as written. How many levels and gates?

(b) Simplify F to a two-level sum of products.

(c) Evaluate F at $A = B = C = D = 0$ in both forms.

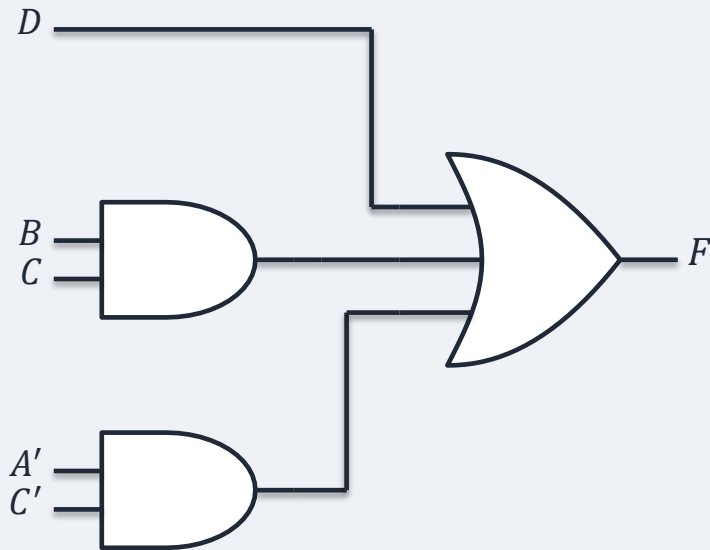
Hint for (b): multiply out $(D + C')(A' + C)$, then use absorption.

Your turn 2 — Solution

§2.8

SOLUTION

(b) $F = D + BC + A'C'$



- 1 (a) Gates: 2 NOT, 2 OR, 2 AND, one 3-input OR
- 2 Longest path OR $\rightarrow$ AND $\rightarrow$ OR: 3 levels
- 3 (b) $(D + C')(A' + C) = A'D + CD + A'C' + CC'$ (distributive)
- 4 $CC' = 0$: $A'D + CD + A'C'$ (complement)
- 5 $F = D + BC + A'D + CD + A'C'$
- 6 $D + A'D = D$ and $D + CD = D$ (absorption)
- 7 $F = D + BC + A'C'$: 2 levels
- 8 (c) As written: $0 + 0 + (0 + 1)(1 + 0) = 1$
- 9 Simplified: $0 + 0 + 1 \cdot 1 = 1$ ✓

Concept check

1

How many minterms does a 4-variable function have?

2

If $F = \Sigma m(0,3,5)$ over 3 variables, what is its Π form?

3

Why is a two-level circuit usually faster than a three-level one?

4

What is a positive-logic NAND gate in negative logic?

Key takeaways

Minterm m_j

1 on row j only

Maxterm M_j

$$M_j = m'_j$$

$$\Sigma \leftrightarrow \Pi$$

complementary index lists

Standard forms

SOP (AND-OR), POS (OR-AND)

Two variables

$$2^{2^n} = 16 \text{ functions}$$

Neg. logic

turns a gate into its dual

Read the canonical form from the truth table; simplify from there.

Homework #2 due at the start of Lecture 5

This afternoon's lab: Lab A — Integrated circuits and data sheets (74LS04)

Next: integrated circuits and logic families